**14** Ayshab walked $x$ miles at 4 mph.
She then walked $2x$ miles at 3 mph.

(a) Find Ayshab's average speed for the whole journey.
Give your answer as a mixed number.

Time = distance ÷ speed

**1 mark**
Find both times.

Total time = $\frac{x}{4} + \frac{2x}{3} = \frac{11x}{12}$

**1 mark**
Total distance ÷ total time.

Average speed = $3x \div \frac{11x}{12}$

**1 mark**
Cancel x.

$= \frac{36}{11} = 3\frac{3}{11}$ mph

**1 mark**
Mixed number.

..... mph

**(4)**

The second part of the journey took 25 minutes longer than the first part of the journey.

(b) Find the value of $x$.

25 minutes = $\frac{25}{60}$ hours

**1 mark**
Use hours throughout.

$\frac{2x}{3} - \frac{x}{4} = \frac{25}{60}$

**1 mark**
Second time minus first.

×60 throughout: $40x - 15x = 25$

**1 mark**
Clear fractions.

$\frac{25x}{\div 25} = \frac{25}{\div 25}$

$x = 1$

**1 mark**
x = 1 mile.

$x =$ .....

**(4)**

**(Total for Question 14 is 8 marks)**